TECHPLUS

2026 Edition

GAME DEVELOPMENT

Complete Learning Roadmap

Create engaging games through mechanics design, engine mastery, and optimized runtime behavior.

- Game loops
- Input systems
- Physics fundamentals
- Object-oriented gameplay code
- Level design principles
- State machines

01 Fundamentals

02 Core Concepts

03 Tools and Frameworks

04 Real Projects

05 Advanced Topics

5 Learning Stages

20+ Core Topics

10+ Projects

100% Game

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Learning Path Overview

Five progressive stages from fundamentals to production mastery

01 Fundamentals

Game loops • Input systems • Physics fundamentals • Object-oriented gameplay code

02 Core Concepts

Level design principles • State machines • Animation systems • Audio integration

03 Tools and Frameworks

Unity or Unreal toolchains

04 Real Projects

2D platformer • 3D action prototype • Multiplayer mini-mode • In-game economy prototype

05 Advanced Topics

Live ops basics • Performance optimization • Cross-platform build strategy • Monetization and retention systems

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Game loops
- Input systems
- Physics fundamentals
- Object-oriented gameplay code

Tools & Frameworks

- Game loops
- Input systems
- Physics fundamentals
- Object-oriented gameplay code

Recommended Projects

- Game Development workshop using Game loops
- Game Development portfolio project with Input systems

Stage 1 Milestone Complete Game Development workshop using Game loops and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Level design principles
- State machines
- Animation systems
- Audio integration

Tools & Frameworks

- Level design principles
- State machines
- Animation systems
- Audio integration

Recommended Projects

- Game Development workshop using Level design principles
- Game Development portfolio project with State machines

Stage 2 Milestone Complete Game Development workshop using Level design principles and document the decisions, trade-offs, and validation checks before moving to Stage 3.

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Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Unity or Unreal toolchains
- Asset pipelines
- Scripting workflows
- Profiling tools

Tools & Frameworks

- Unity or Unreal toolchains

Recommended Projects

- Game Development workshop using Unity or Unreal toolchains
- Game Development portfolio project with Asset pipelines

Stage 3 Milestone Complete Game Development workshop using Unity or Unreal toolchains and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- 2D platformer
- 3D action prototype
- Multiplayer mini-mode
- In-game economy prototype

Tools & Frameworks

- 2D platformer
- 3D action prototype
- Multiplayer mini-mode
- In-game economy prototype

Recommended Projects

- Game Development workshop using 2D platformer
- Game Development portfolio project with 3D action prototype

Stage 4 Milestone Complete Game Development workshop using 2D platformer and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

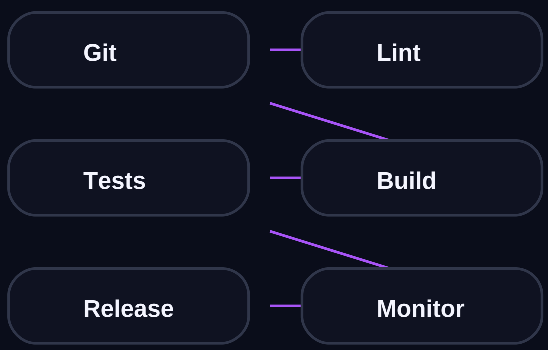
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Live ops basics
- Performance optimization
- Cross-platform build strategy
- Monetization and retention systems

CI/CD Pipeline



Stage 5 Milestone

A production-grade game development workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

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Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE Game Development docs and community guides

Link: https://learn.unity.com/

FREE Build one portfolio project per stage before moving ahead

Link: https://roadmap.sh/game-developer

PAID Structured specialization track focused on Game Development

Link: https://roadmap.sh/game-developer

FREE Game Development roadmap reference

Link: https://roadmap.sh/game-developer

FREE Official documentation

Link: https://learn.unity.com/

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Game Development?
- Which project best proves Stage 4 competency in Game Development?
- What trade-offs appear in Stage 5 decisions for Game Development?